Mohammadreza Narimani

Artificial Intelligence | Computer Vision | Digital Agriculture | Precision Sensing Systems

PhD Candidate in Biological Systems Engineering

University of California, Davis

Biological and Agricultural Engineering Department

California, US

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Research Interests

Remote Sensing of Agriculture and Environment Earth Observation Digital Agriculture Machine Learning Computer Vision Precision Agriculture Hyperspectral Imaging LiDAR Satellite Data Analysis IoT Systems Plant Phenotyping 3D Radiative Transfer Modeling Wildfire Detection

Education

Ph.D. Candidate of Biological Systems Engineering 2022–Present

University of California, Davis, US **GPA: 4.0/4.0**
Thesis: AI-Driven Remote Sensing and Spectral Modeling for Advanced Monitoring of Tomato Crops
Advisors: Prof. Alireza Pourreza, Prof. Ali Moghimi

M.Sc. of Biosystem Mechanical Engineering 2019–2021

University of Tehran, Iran **GPA: 4.0/4.0 (First Rank)**
Thesis: Designing and Manufacturing an Experimental Smart Greenhouse for High-Throughput Stress Phenotyping and Plant Disease Detection by Implementing Internet of Things System and Machine Learning

B.Sc. of Biosystem Mechanical Engineering 2015–2019

University of Tehran, Iran **GPA: 4.0/4.0 (First Rank)**
Thesis: Design and construction of electric turning machine for urban agriculture

Research Experience

Research Assistant 2022–Present

Digital Agriculture Laboratory, University of California, Davis

- AI-Driven Remote Sensing and Spectral Modeling for Advanced Monitoring of Tomato Crops
- 3D Radiative Transfer Modeling for Specialty Crops

- Wildfire Detection and Monitoring Using Satellite Imagery and Deep Learning
- Image semantic segmentation using FCN_AlexNet, SegNet, and U-Net for Agricultural Vision Competition

Research Assistant 2016–2022

University of Tehran, Iran

- High-Throughput Wheat Phenotyping for Agricultural Research Institute of Iran, Wheat Species Bank
- Detecting fungal disease of lettuce with thermal camera and CNNs
- Image processing algorithms for measuring friction coefficient and angle of repose in rice grain
- IoT device development using Arduino, Raspberry Pi, Ethernet, and ESP8266
- Designing novel fogponic and centrifugal irrigation systems for aeroponic greenhouses
- Deep learning algorithms for detecting powdery tomato disease
- Designing strawberry harvester robot

Internship 2017–2022

Agricultural Engineering Research Institute of Iran (AERI)

Technical Skills

Remote Sensing Tools: Sentinel-2, Landsat-9, EMIT, ECOSTRESS, DJI Phantom 4 Multispectral, Aerial Pika-L Hyperspectral Sensor, Aerial Phoenix LiDAR Systems

Programming Languages: Python, R, MATLAB, C++

Amazon Web Services: Lambda, S3, App Runner, API Gateway, EC2, Elastic Container Registry, SageMaker, Bedrock, IAM, CloudFormation, CloudWatch

Techniques: Computer Vision, Image Processing, Web App Development, Internet of Things, Robotics

Prominent Libraries: Keras, TensorFlow, PyTorch, scikit-learn, geemap, rasterio, gdal, pandas, NumPy, OpenCV, PlantCV, scikit-image

GIS: ArcGIS, QGIS, Google Earth Engine, ENVI, Pix4D, Resonon, Spatial Explorer, Microstation, Terrasolid

Computer-Aided Design and Engineering: CATIA, SolidWorks, AutoCAD, Abaqus, Ansys, Keyshot

Artificial Intelligence

Deep Learning: Instance segmentation (Mask R-CNN, YOLO, SAM); semantic segmentation of images (U-Net, FCN, SegNet, DeepLab); image classification and object detection; frameworks: PyTorch, TensorFlow, Keras

Machine Learning: Regression (linear, polynomial, ridge, LASSO, Gaussian process); classification (Random Forest, SVM, XGBoost, logistic regression, k-NN, decision trees, gradient boosting); model selection and validation; scikit-learn

Large Language Models (LLMs): GPT-4, Claude (Anthropic), Llama (Meta), Amazon Bedrock, Kiro; prompt engineering, RAG (retrieval-augmented generation), API integration and deployment

Journal Publications

1. Mitra, S., **Narimani, M.R.**, Farajpoor, P. (2026). A Sentinel-2 Benchmark Dataset for Deep-Learning Active-Fire Segmentation Across 25 California Wildfires, July 2020–August 2026. *Data in Brief*, 113357. DOI Link
2. Farajpoor, P., **Narimani, M.R.** (2026). Explainable GeoAI for Urban Wildfire Structure Loss: Vegetation and Built Environment in the 2025 Palisades Fire. *Frontiers in Sustainability*, 7, 1975724. DOI Link
3. Farajpoor, P., **Narimani, M.R.** (2026). From Wildfire Severity to Snow Persistence: A Multisource GeoAI Study of the 2020 Creek Fire. *arXiv preprint* arXiv:2609.36345. DOI Link
4. Farajpoor, P., Ardebili Pour, M., Ghiasi, M.B., **Narimani, M.R.** (2026). Multisource Remote Sensing and Geospatial Analysis of Vineyard Wildfire Impacts and Exposure: The 2019 Kincade Fire. *Frontiers in Sustainability*, 7, 1997548. DOI Link
5. **Narimani, M.R.**, Mitra, S., Farajpoor, P. (2026). From crown candidates to neighborhood screening: integrating optical GeoAI and spatial modeling for urban-canopy assessment in Davis, California. *arXiv preprint.* DOI Link
6. **Narimani, M.R.**, Anand, V., Farajpoor, P. (2026). Farmland Extent and Visible Boundary Mapping from 1 m NAIP Imagery Using Residual U-Net and Text-Prompted SAM 3 Refinement. *arXiv preprint.* DOI Link
7. **Narimani, M.R.**, Pourreza, A., Farajpoor, P. (2026). Mapping Tomato Cropping Systems in California Using AlphaEarth Geospatial Embeddings and Deep Learning Analysis. *2026 ASABE Annual International Meeting*, Paper No. 2600545. DOI Link
8. **Narimani, M.R.**, Pourreza, A., Moghimi, A., Farajpoor, P. (2026). Sentinel-2 for Crop Yield Estimation: A Systematic Review. *Smart Agricultural Technology*, 14, 102405. DOI Link
9. Farajpoor, P., Pourreza, A., **Narimani, M.R.**, ElKereamy, A., Fidelibus, M. (2025). Multi-Trait Spectral Modeling for Estimating Grapevine Leaf Traits and Nutrients. *Plant Phenomics.* DOI Link
10. **Narimani, M.R.**, Pourreza, A., Moghimi, A., Farajpoor, P., Jafarbiglu, H., Mesgaran, B. (2025). Early Detection of Branched Broomrape (Phelipanche Ramosa) Infestation in Tomato Crops by Using Leaf Spectral Analysis and Machine Learning. *IFAC-PapersOnLine.* DOI Link
11. Alimardani, R., Adedeji, A.A., **Narimani, M.R.** (2025). A Review of IoT Applications in Food Processing and Related Fields. *Journal of Food Processing and Preservation.* DOI Link

12. **Narimani, M.R.**, Pourreza, A., Moghimi, A., Farajpoor, P., Jafarbiglu, H., Mesgaran, B. (2025). Branched broomrape detection in tomato farms using satellite imagery and time-series analysis. *SPIE Defense + Commercial Sensing*. DOI Link
13. Farajpoor, P., Pourreza, A., **Narimani, M.R.**, ElKereamy, A., Fidelibus, M. (2025). Leaf spectral reflectance prediction using multihead attention neural networks. *SPIE Defense + Commercial Sensing*. DOI Link
14. **Narimani, M.R.**, Pourreza, A., Moghimi, A., Mesgaran, B., Farajpoor, P., Jafarbiglu, H. (2024). Drone-based multispectral imaging and deep learning for timely detection of branched broomrape in tomato farms. *SPIE Defense + Commercial Sensing*. DOI Link
15. Akpenpuun, T.D., Ogunlowo, Q.O., Na, W.H., Dutta, P., Rabiu, A., Adesanya, M.A., **Narimani, M.R.**, et al. (2024). Dynamic neural network modeling of thermal environments of two adjacent single-span greenhouses. *Journal of Agricultural Engineering*. DOI Link
16. Jafarbiglu, H., Pourreza, A., **Narimani, M.R.** (2024). Adaptive Canopy Segmentation Using Seam Detection in LiDAR-Derived Surface Models. *SSRN preprint*. Publication Link
17. **Narimani, M.R.**, Hajiahmad, A., Moghimi, A., Alimardani, R., Rafiee, S., Mirzabe, A.H. (2021). Developing an aeroponic smart experimental greenhouse for controlling irrigation and plant disease detection using deep learning and IoT. *ASABE Annual International Meeting*, Paper No. 2101252. Publication Link
18. Elyasi, M., Kianmehr, M.H., **Narimani, M.R.**, Amiri, H. (2018). Designing and Manufacturing Electrical Turner for Urban Agriculture. *First National Conference of Urban Agriculture*, 1(1), 247–261. Publication Link
19. Moghadam, S., Alimardani, R., **Narimani, M.R.**, Moghimi, A. (2026). Internet of Things and Artificial Intelligence for smart agriculture: a systematic mapping review of sensors, platforms, connectivity, and machine learning applications (2014-2024). *AgriRxiv*. DOI Link

Conference Presentations

1. **Narimani, M.R.**, et al. (2025). Early Detection of Branched Broomrape Infestation in Tomato Crops. *8th IFAC Conference on Sensing, Control and Automation Technologies for Agriculture*. Conference Link
2. **Narimani, M.R.**, et al. (2024). Drone-based multispectral imaging and deep learning for timely detection of branched broomrape. *SPIE Defense + Commercial Sensing*. Presentation Link
3. **Narimani, M.R.**, et al. (2024). Early Detection of Branched Broomrape in Tomato by Hyperspectral Sensing. *ASABE 2024*.
4. Jafarbiglu, H., Pourreza, A., **Narimani, M.R.**, Sanden, B., Marino, G., Culumber, M., Mehata, M., Ferguson, L. (2024). Determining the best irrigation strategy for pistachio orchards with saline water and soil. *SPIE Defense + Commercial Sensing*. Presentation Link
5. **Narimani, M.R.**, et al. (2024). Remote Sensing of Branched Broomrape in Tomato. *65th Weed Day, UC Davis*. Event Link
6. **Narimani, M.R.**, et al. (2023). Wildfire Detection and Monitoring Using Satellite Imagery and Deep Learning. *ASABE 2023*. Schedule Link
7. Jafarbiglu, H., Pourreza, A., **Narimani, M.R.**, Williams, D., Ferguson, L. (2023). Determining the best irrigation practices for pistachio orchards with saline water and soil. *ASABE 2023*. Schedule Link
8. **Narimani, M.R.**, Hajiahmad, A., Moghimi, A., Alimardani, R., Rafiee, S., Mirzabe, A.H. (2021). Developing an aeroponic smart experimental greenhouse for controlling irrigation and plant disease detection using deep learning and IoT. *ASABE 2021*.

Conference Posters

1. **Narimani, M.R.**, Pourreza, A., Farajpoor, P. (2026). Mapping Tomato Cropping Systems in California Using AlphaEarth Geospatial Embeddings and Deep Learning Analysis. *ASABE Annual International Meeting, Indianapolis, IN.*
2. Farajpoor, P., Pourreza, A., **Narimani, M.R.** (2025). Domain Adaptation Approaches to Improve Leaf Trait Estimation by Reducing Error Propagation in Hybrid Modeling. *AGU 2025.*
3. **Narimani, M.R.**, Pourreza, A., Moghimi, A., Farajpoor, P., Jafarbiglu, H., Mesgaran, B. (2025). Branched broomrape detection in tomato farms using satellite imagery and time-series analysis. *SPIE Defense + Commercial Sensing.* Poster Link
4. Jafarbiglu, H., Pourreza, A., **Narimani, M.R.**, Bailey, J., Taylor, G. (2024). Seam Carving for Tree Segmentation in Dense Tree Farms. *ASABE 2024.*
5. **Narimani, M.R.**, Hajiahmad, A., Moghimi, A., Alimardani, R., Rafiee, S., Mirzabe, A.H. (2021). Developing an aeroponic smart experimental greenhouse for controlling irrigation and plant disease detection using deep learning and IoT. *ASABE 2021.*

Teaching Experience

Teaching Assistant, University of California, Davis 2022–Present

- Applied Statistics in Agricultural Sciences Lab Materials
- Environmental Analysis Using GIS Lab Materials
- Environmental Monitoring
- Introduction to Unmanned Aerial Systems for Agriculture & Environmental Science
- Communications and Computing Technology (IoT in Ag) Lab Materials

Teaching Assistant, University of Tehran, Iran 2016–2022

- Programming with Python and MATLAB
- Computer Vision and Artificial Intelligence
- Industrial drawing with CATIA, SolidWorks and AutoCAD

Teaching Physics, Bonyad Amoozeshi Ghalamchi 2014–2016

Awards and Honors

- First Rank, Excellence in Specialty Crops Award, Farm Robotics Challenge (Selected from 96 global teams) 2026
- Ernest E. Hill Memorial Graduate Fellowship, UC Davis 2026–2027
- Walter Rosenberg Research Fellowship (GBSE Fellowship), UC Davis 2026
- Future Undergraduate Science Educators (FUSE) Program Scholarship 2025–Present
- Best Presentation Award, ASABE Annual International Meeting 2024
- Agricultural Genome to Phenome Initiative Scholarship 2024
- Bill And Jane Fischer Vegetation Management Scholarship 2024
- Member of SPIE Defense + Commercial Sensing 2024–Present
- Peter J. Shields and Henry A. Jastro Research Award 2023
- Dean's Distinguished Graduate Fellowship 2022

- Member of ASABE (American Society of Agricultural and Biological Engineers) 2021–Present
- First rank prize, Iran IoT and Computer Vision Competition 2020

Leadership and Service

Leadership Positions

- Graduate Student Leader, UC Davis Team, Farm Robotic Challenge, UCANR Innovate & AIFS 2025–Present
- Vice President, Graduate Student Association, BAE Department, UC Davis 2024–Present
- Representative, Graduate Student Association, BAE Department, UC Davis 2023–2024
- Election Committee Member, Graduate Student Association, BAE Department 2025

Conference Service

- Session Moderator, Session 328: "Imaging Spectroscopy and Multimodal AI Systems for Automated Crop Quality Evaluation," ASABE Annual International Meeting, Indianapolis, IN July 15, 2026

Mentorship

- Mentor, E-SEARCH Program, UC Davis (4 undergraduate students) 2024–Present
- Mentor, AgTech Workshop (Drone & IoT), AI Institute for Next Generation Food Systems 2023–Present
- Mentor, Young Scholar Program, UC Davis (high school students) 2023

Mentees' Research Posters

- Sun, V.J., **Narimani, M.R.** Deep Learning-Based Snow Monitoring in California Using Sentinel-2 Satellite Data. ESEARCH Summer 2025.
- Kumar, S., **Narimani, M.R.**, et al. Enhancing Orchard Management with Deep Learning: Tree Segmentation Using Geospatial SAM2 Model. ESEARCH Fall 2024.
- Richmond, N., **Narimani, M.R.**, et al. Global Vegetation and Climate Insights Portal (GVCIP). ESEARCH Summer 2024.
- Tran, Q., **Narimani, M.R.**, et al. Enhancing Wildfire Monitoring Through Remote Sensing. ESEARCH Spring 2024.

Editorial and Peer Review Activities

Reviewer for the following journals:

- Weed Research (Wiley) 2026–Present
- Scientific Reports (Springer Nature) 2024–Present
- Journal of Agriculture and Food Research (Elsevier) 2024–Present
- Journal of Food Science (Wiley) 2024–Present
- Smart Agricultural Technology (Elsevier) 2024–Present
- Computers and Electronics in Agriculture (Elsevier) 2024–Present
- Information Processing in Agriculture (Elsevier) 2024–Present
- Remote Sensing Applications: Society and Environment (Elsevier) 2024–Present
- NJAS: Impact in Agricultural and Life Sciences (Taylor & Francis) 2024–Present

Media Coverage

Parasitic Weeds Threaten Tomato Plants on California Farms - Featured in:

- UC Davis Newsletter Read Article
- UC Davis College of Agricultural and Environmental Sciences Read Article
- Daily Democrat Newspaper Read Article
- Ag Alert Magazine Read Article
- Capital Press Newsletter Read Article
- Organic Growers Newsletter Read Article
- UC Davis Plant Sciences Newsletter Read Article
- Tomato News Read Article
- California Fruit and Vegetable Read Article

First Rank, Excellence in Specialty Crops Award, Farm Robotics Challenge 2026 - Featured in:

- UCANR News Release (Awards at USA Plug and Play Summit) Read Article
- Farm Robotics Challenge Official 2026 Results View Results
- Project BloomSense Official Video Entry Watch Video

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